1

2

### What is voltage?

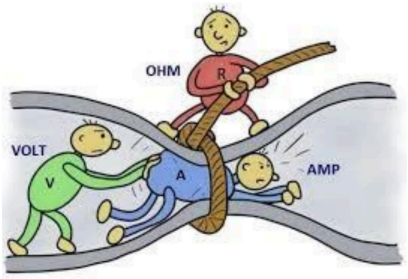
Voltage, often measured in volts (V), is the electrical potential difference between two points in a circuit. It represents the force that drives electric current.

### What is current?

Current, measured in amperes (A), is the flow of electric charge in a circuit. It is the rate at which electric charges (usually electrons) move through a conductor.

### What is resistance?

Resistance, measured in ohms (ΩΩ, is the opposition that a material offers to the flow of electric current. It depends on the material's properties, length, cross-sectional area, and temperature.




### How are voltage, current, and resistance related?

They are related by Ohm's Law, which states that voltage (V) is equal to the product of current (I) and resistance (R) in a circuit: ⁢=⁢×⁢V=I×R. This relationship helps to understand how changes in one of these parameters affect the others.

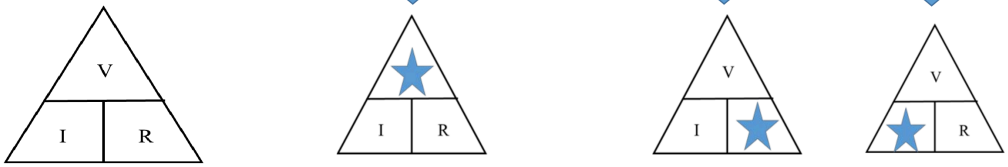
3

## Ohm's Law

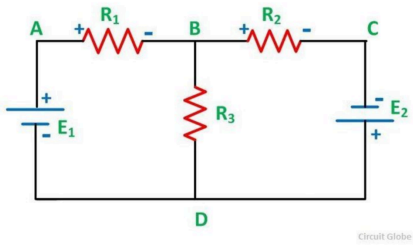
Ohm's law states that the current (I) flowing through a circuit is directly proportional to the potential difference (V) across the circuit and inversely proportional to the resistance (R) of the circuit provided the temperature remains constant.

Mathematically, I=V/R ..... (1)
V=IR..... (2)
R=V/I .....(3)

### The magic triangle



7




**A node** is a point in the network or circuit where two or more circuit elements are joined. For example, in the above circuit diagram, A and B is the node points.

**A junction** is a point in the network where three or more circuit elements are joined. It is a point where the current is divided. In the above circuit, B and D are the junctions.

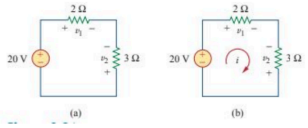
**Branch** is the part of a network, which lies between the two junction points is called a Branch. In the above circuit DAB, BCD and BD are the branches of the circuit.

A closed path of a network is called a **loop**. ABDA, BCDB are loops in the above circuit diagram shown.

The most elementary form of a loop which cannot be further divided is called a **mesh.**

8

For the circuit in Fig. 2.21(a), find voltages $v_1$ and $v_2$.




Find $v_1$ and $v_2$ in the circuit of Fig. 2.22.

**Answer:** 12 V, −6 V.

**Practice Problem 2.5**

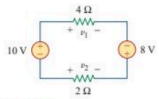



**Figure 2.22**
For Practice Prob. 2.5.

**Example 2.7** Find current $i_o$ and voltage $v_o$ in the circuit shown in Fig. 2.25.

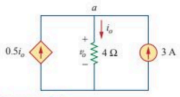



**Figure 2.25**
For Example 2.7.

**Solution:**
Applying KCL to node $a$, we obtain

$$3 + 0.5i_o = i_o \quad \Rightarrow \quad i_o = 6 \text{ A}$$

For the 4-Ω resistor, Ohm's law gives

$$v_o = 4i_o = 24 \text{ V}$$

9

**Practice Problem 2.7** Find $v_o$ and $i_o$ in the circuit of Fig. 2.26.

**Answer:** 8 V, 4 A.

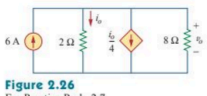



**Figure 2.26**
For Practice Prob. 2.7.

**Example 2.8** Find currents and voltages in the circuit shown in Fig. 2.27(a).

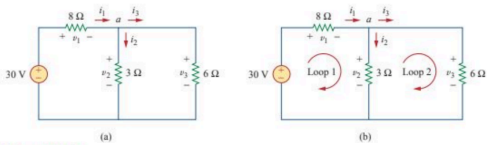



**Figure 2.27**
For Example 2.8.